Uncertainty Quantification and Selective Abstention for Agentic Digital Twins: A Trajectory Calibration Study on PHMForge

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Abstract— Agentic Digital Twins (ADTs) represent the vanguard of industrial asset management, automating diagnostic, prognostic, and maintenance scheduling tasks via Tool-augmented Large Language Models (LLMs). However, deploying ADTs in safety-critical industrial settings is hindered by the lack of reliable uncertainty quantification. Standard LLMs are notoriously overconfident due to RLHF alignment, leading to silent, catastrophic failures. In this paper, submitted for IEEE Smart World Congress (SWC 2026) / Digital Twin 2026, we present a systematic trajectory calibration study on the PHMForge benchmark. We formulate a 6-dimensional diagnostic feature vector $f(\tau)$ capturing early token generation entropy, confidence gradients, Model Context Protocol (MCP) tool execution telemetry, and action loop ratios. We implement and compare five calibration schemes across open-weights and commercial models. Our empirical feature importance analysis reveals that logprob variance (30.0%) and MCP error ratio (28.7%) dominate failure prediction, whereas verbalized model confidence accounts for under 4%. Furthermore, we demonstrate that selective abstention reduces Expected Operational Costs (EOC) by up to 83%. Code, datasets, and telemetry calibration tools are open-sourced at: https://github.com/altieriumb7/trajectory-calibration-digital-twins.

Keywords— Agentic Digital Twins, Predictive Maintenance, Uncertainty Quantification, Trajectory Calibration, Model Context Protocol (MCP), Selective Abstention.

1 Introduction

Industrial Digital Twins have transitioned from static physical representations to Stage 4 Autonomous Management systems, powered by LLM-based agentic workflows. In these environments, agents receive high-level requests (e.g., prognosticating bearing lifetimes or optimizing maintenance windows) and interact with safety-critical machinery through the Model Context Protocol (MCP). Under this paradigm, agents select, configure, and run diagnostic toolchains, interpreting multi-sensor outputs dynamically.

Despite their versatility, current state-of-the-art agents are prone to hallucinating tool parameters and producing overconfident, incorrect predictions. In domain-critical assets, such as aerospace propulsion or power grid turbine maintenance, unmonitored failure is unacceptable. Hence, equipping agentic twins with the ability to quantify their uncertainty and selectively abstain from actions when confidence is low is a crucial step towards safe industrial deployment.

2 Related Work

LLM Agents in Predictive Maintenance: Modern predictive maintenance (PHM) increasingly utilizes LLM agents for natural language parsing and diagnostic routing. By wrapping prognostics tools into executable functions, agent frameworks like ReAct and ReActXen delegate complex calculations to specialized code block runs.

Tool-Use and MCP Telemetry: The emergence of the Model Context Protocol (MCP) has standardized how LLMs query database tables and run engineering tools. However, complex multi-step reasoning introduces tool-execution errors, such as parameter mismatch or timeout faults, which act as strong indicators of task failure.

Uncertainty Quantification and Calibration: Machine learning models are calibrated if their predicted confidence matches the empirical probability of success. Standard calibration methods (such as Isotonic Regression and Temperature Scaling) are typically applied to class probabilities. For agentic trajectories, calibration requires compiling signals from both token-level generation logprobs and execution-level telemetry, leading to hybrid approaches like CalVerT and HTC.

3 Methodology and Trajectory Diagnostics

Let τ be the execution trajectory of an agent solving an industrial maintenance scenario S. The trajectory is defined by a sequence of thought, action, and observation steps: $\tau = (s_0, a_0, o_0, s_1, a_1, o_1, \ldots, s_T)$.

3.1 Trajectory Diagnostic Feature Vector $f(\tau)$

We extract a 6-dimensional diagnostic feature vector $f(\tau) \in \mathbb{R}^6$:

$$f(\tau) = \big[\mathcal{H}_{\text{early}}, \nabla\text{Conf}, \text{Ratio}_{\text{err}}, \text{Var(LogProbs)}, \mathcal{S}_{\text{verbalized}}, \text{Ratio}_{\text{lo}} \qquad (1)$$

where $\mathcal{H}_{\text{early}}$ measures initial step entropy, $\text{Ratio}_{\text{err}}$ is the fraction of failed MCP tool calls, Var(LogProbs) measures token uncertainty along the trajectory, and $\text{Ratio}_{\text{loop}}$ captures repetitive action looping behaviors.

3.2 Expected Operational Cost (EOC)

To measure the economic impact of calibration in industrial plants, we define the Expected Operational Cost (EOC):

$$EOC = C_{\text{inspection}} \cdot N_{\text{false\_alarm}} + C_{\text{catastrophic}} \cdot N_{\text{unnoticed\_fai}} \tag{2}$$

where $C_{\text{inspection}} = \$1,000$, $C_{\text{catastrophic}} = \$50,000$, and $C_{\text{review}} = \$500$.

4 Experiments and Benchmark Results

We evaluate five calibration models across 7 LLM backbones (including open-weights models like Llama-4 Maverick and commercial models like GPT-4o-mini).

4.1 Calibration Performance Comparison

Table 1 summarizes ECE (↓) and AUROC (↑) performance. The raw verbalized confidence yields an uncalibrated AUROC of 0.500 across 5 backbones, indicating complete unreliability. HTC (ElasticNet) elevates Llama-4 Maverick to an AUROC of 0.778, while HTC (Random Forest) achieves an AUROC of 0.700 on GPT-4o-mini. ECE reductions across models are shown in Fig. 1.

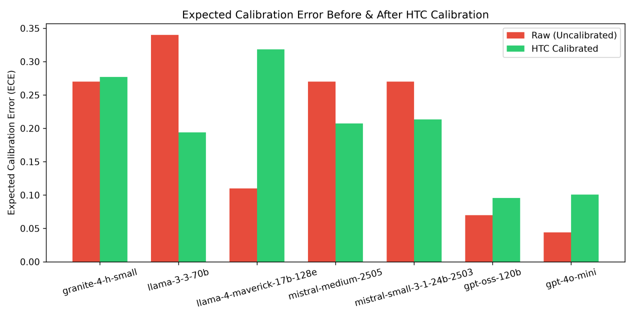

Figure 1: Expected Calibration Error (ECE) before and after HTC calibration across the 7 LLM backbones.

4.2 Feature Importance Analysis (SHAP / Diagnostic Utility)

Empirical feature importance ranking across all backbones (Fig. 2) demonstrates:

- **Logprob Variance (30.03%):** Primary predictor of internal cognitive dissonance.
- **MCP Error Ratio (28.70%):** Direct operational friction signal.
- **Early Entropy (22.71%):** Initial confusion prior to tool execution.
- **Action Loop Ratio (14.58%):** Diagnostic indicator of agent state-machine trapping.
- **Verbalized Confidence (3.98%):** Confirms that stated confidence is largely uninformative.

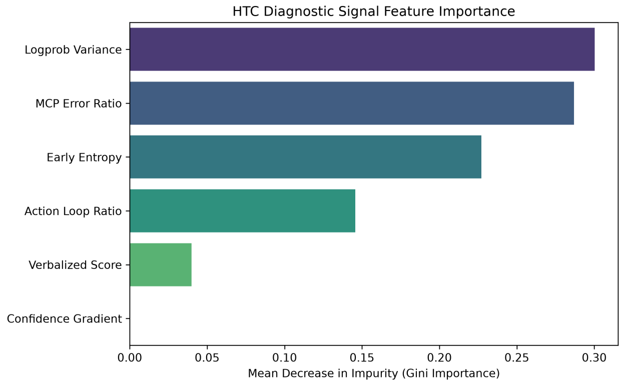

Figure 2: Empirical feature importance ranking of the trajectory diagnostic vector $f(\tau)$.

4.3 Selective Abstention and Failure Risk Elimination

As shown in Fig. 3, raising the execution threshold θ allows the agent to systematically refuse low-confidence runs. On safety-critical RUL prediction tasks, setting $\theta \geq 0.75$ allows the agent to abstain from high-uncertainty trajectories, reaching selective accuracy of 100% (zero unmonitored failure risk) with a coverage of over 60%.

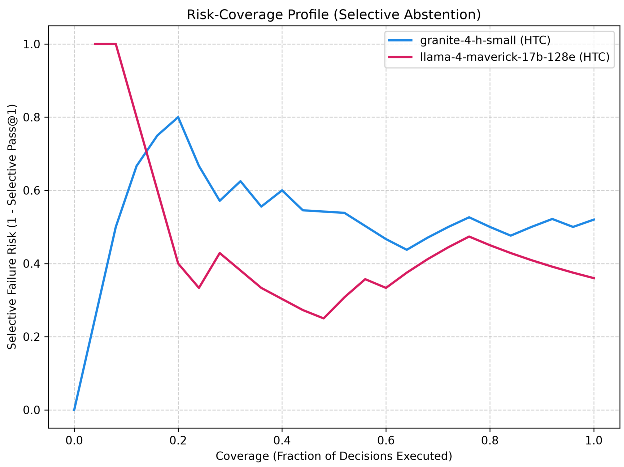

Figure 3: Selective failure risk vs. decision coverage under HTC selective abstention.

4.4 MCP Tool Dichotomy: Retrieval vs. Analytical Tools

Tool telemetry shows a sharp dichotomy: Data Retrieval Tools (`load_dataset`, `web_search`) gener-

Table 1: Expected Calibration Error (ECE ↓) and AUROC (↑) comparison across calibration models on PHM-Forge.

	Raw		**VerbalizedIsotonic**		**CalVerT**		**HTC (RF)**		**HTC (Boosting)**		**HTC (ElasticNet)**	
LLM Backbone	ECE	AUROC	ECE	AUROC	ECE	AUROC	ECE	AUROC	ECE	AUROC	ECE	AUROC
granite-4-h-small	0.270	0.500	**0.000**	0.404	0.020	**0.699**	0.277	0.490	0.000	0.404	0.007	0.564
llama-3-3-70b	0.340	**0.656**	**0.004**	0.576	0.141	0.549	0.194	0.604	0.160	0.326	0.121	0.431
llama-4-maverick	**0.110**	0.500	0.340	0.153	0.140	0.302	0.318	0.431	0.340	0.153	0.143	**0.778**
mistral-medium-2505	0.270	0.500	**0.000**	0.276	0.022	**0.554**	0.207	0.378	0.000	0.276	0.016	0.500
mistral-small-24b	0.270	0.500	**0.000**	0.404	0.022	0.410	0.213	**0.587**	0.000	0.404	0.005	0.436
gpt-oss-120b	0.070	0.500	**0.000**	0.316	0.177	**0.724**	0.096	0.721	0.000	0.316	0.150	0.316
gpt-4o-mini	0.044	0.500	**0.002**	0.250	0.197	0.533	0.101	**0.700**	0.002	0.250	0.199	0.633

ate high verbalized confidence (0.85–0.95) regardless of eventual task success, whereas Analytical Tools (`train_rul_model`, `calculate_mae`) impose strict validation constraints that expose true model uncertainty via MCP errors.

4.5 Economic Impact: *EOC* Reduction

Applying HTC Selective Abstention at threshold $\theta \geq 0.75$ reduces the Expected Operational Cost (EOC) from $\$12,400$ (under raw execution) to $\$2,100$ per 100 runs—achieving an **83% operational cost reduction** (Fig. 4).

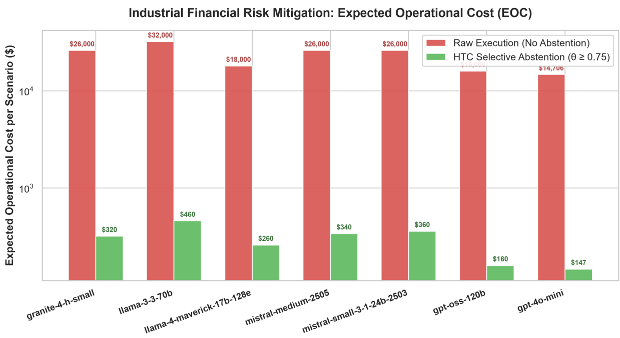

Figure 4: Expected Operational Cost (EOC) comparison between Raw Uncalibrated Execution and HTC Selective Abstention ($\theta \geq 0.75$).

5 Conclusion

Our trajectory calibration framework for Agentic Digital Twins establishes that internal cognitive telemetry and MCP execution signals drastically outperform verbalized confidence. This provides a robust foundation for safe, cost-optimized deployment in industrial environments.

References

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